LALITH VARA PRASAD KARAMPUDI

Software Engineer — Python, Backend Services, Docker & Testing

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PROFESSIONAL SUMMARY

Recent Computer Science graduate (B.Tech, Data Science specialization) who designs, tests, deploys and maintains backend software across the full development lifecycle: Python services with validated REST APIs (FastAPI), Docker and docker-compose deployments, automated unit and integration tests, and GitHub Actions CI. Applies clean interfaces (pluggable model and storage layers), defensive design (input validation, step and loop limits, resumable caches) and clear documentation. Works independently on multi-part projects and manages own priorities. Published researcher (ICACTEA-2025); Oracle Foundations Associate certified.

TECHNICAL SKILLS

Programming: Python (primary), SQL, Java and C++ (coursework, OOP), Data Structures & Algorithms (coursework), Git/GitHub

Backend & Deployment: Distributed Queue Fundamentals (leases, retries, idempotency, dead-lettering), Redis, RESTful APIs (FastAPI, pydantic validation), Docker, docker-compose, GitHub Actions CI/CD, Deployment on Render, Vector Database (ChromaDB)

Engineering Practice: SDLC (design, implement, test, deploy, document), pytest (unit/integration, mocking), Debugging and Root-Cause Analysis, Defensive Design, Failure/Chaos Testing, Code Documentation

ML / AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, RAG, LLM Integrations, AI Agents, MLflow

Certifications: Oracle Foundations Associate

PROJECTS

Task Pipeline — Two-Service Distributed Job Queue | Python, FastAPI, Redis, Docker Compose, pytest, GitHub Actions | github.com/lalith0087/task-pipeline

- Built a two-service system (FastAPI API + horizontally scalable worker service) on a Redis queue, with separate Docker images and a Docker Compose setup.
- Engineered reliability: atomic Lua-script dequeue for competing consumers, leases with heartbeats, fencing tokens against stale workers, retries with exponential backoff and jitter, dead-letter list, idempotency keys and graceful SIGTERM shutdown.
- Verified with 21 tests against a real Redis and a manual kill -9 test (a crashed worker's job was recovered by another worker); CI runs the tests and a Compose smoke test.

Guarded RAG Agent — Tool-Calling Agent with Guardrails | Python, FastAPI, Ollama, ChromaDB, pydantic, pytest, GitHub Actions | github.com/lalith0087/rag-agent-guardrails

- Built a framework-free AI agent with schema-validated tool calling, served through a FastAPI REST endpoint that returns the full step trace.
- Designed input/output validation guardrails (injection screening, argument validation, grounding checks) with 31 offline tests and CI; Dockerized with docker-compose.

Telco Customer Churn — End-to-End ML Pipeline with MLflow | Python, SQL, Scikit-learn, MLflow, FastAPI, Docker, Ollama | github.com/lalith0087/telco-churn-mlops

- Designed and deployed a tuned XGBoost model (ROC-AUC 0.84) to predict customer churn, covering data preparation, EDA, and feature engineering across 75+ MLflow/Optuna-tracked experiments.
- Added LLM-powered explainability endpoints translating model outputs and feature contributions into clear insights for non-technical stakeholders.

Fraud Detection MLOps | Python, SQL, Scikit-learn, MLflow, FastAPI, Docker, GitHub Actions | github.com/lalith0087/fraud-detection-mlops

- Worked with a large, complex real-world dataset (284,807 transactions, 0.17% fraud) to evaluate and recommend a business strategy: trained a class-weighted gradient-boosting model reaching PR-AUC 0.735 / ROC-AUC 0.962 on a stratified held-out set, selecting PR-AUC over accuracy due to the extreme class imbalance.
- Identified a gap in F1-based thresholding and built a cost-based alternative (configurable missed-fraud vs. false-alarm cost): cut expected cost per 1,000 transactions by ~14% versus the F1-optimal threshold (34.15 to 29.23) while holding recall at 0.88.
- Tracked experiments in MLflow and built a model registry with gated promotion that blocks candidates with lower PR-AUC.
- Added PSI-based data-drift monitoring to continuously assess data quality and flag inconsistencies in incoming data versus the training distribution.
- Delivered the model via a validated FastAPI service (tunable threshold) in Docker, with GitHub Actions CI (6 automated tests).

Local RAG Assistant — Generative AI / NLP System | Python, ChromaDB, Ollama, FastAPI | github.com/lalith0087/local-rag-assistant

- Designed and built a fully local Retrieval-Augmented Generation (LLM/GenAI) system from first principles — embeddings, vector retrieval, and context synthesis — answering natural-language questions grounded in source text with exact citations.
- Applied NLP techniques via structure-aware text chunking, improving retrieval relevance over naive slicing, and implemented prompt engineering with a model-agnostic LLM integration layer.
- Evaluated output quality, traced a hallucination failure mode to its root cause and documented a mitigation; 21 tests and CI.

Second Brain GraphRAG — Local Knowledge-Graph Assistant | Python, Ollama, ChromaDB, NetworkX, FastAPI, pytest, GitHub Actions | github.com/lalith0087/second-brain-graphrag

- Built a fully local, private GraphRAG system: schema-validated LLM extraction of entities and relations (with retries) into a NetworkX knowledge graph alongside a ChromaDB index, answering questions through entity linking and 2-hop graph expansion with cited facts.
- Designed a benchmark harness comparing GraphRAG with plain RAG on single- vs multi-hop questions over 62 chunks (198 entities, 165 relations). Found no accuracy gain at 1B-parameter scale (both 6/16), tested and ruled out prompt truncation as the cause, and published the negative result with untested hypotheses.

Image Captioning using CNN + LSTM | Python, TensorFlow, PyTorch | Academic research project (ICACTEA-2025)

- Led a team through architecture design, data preprocessing, training, and evaluation for this computer vision + NLP research project; co-authored the resulting paper, published at ICACTEA-2025, Aditya University (April 2025).
- Later re-implemented the model in PyTorch with a reproducible BLEU evaluation: BLEU-1 0.498 / BLEU-4 0.128 on a held-out Flickr8k split after 20 epochs, with a split-fingerprint check to prevent leakage, 7 tests and GitHub Actions CI.

PUBLICATION

"Image Captioning using CNN and LSTM," ICACTEA-2025, Aditya University — April 2025.

EDUCATION

B.E. / B.Tech — Computer Science (Data Science) | Vignan Institute of Technology and Science | 2020–2025 | CGPA 6.9/10

Coursework: Machine Learning, Statistical Methods, Data Structures, Python, Data Science Libraries, OOP.

Intermediate (MPC), Narayana Junior College, 2018–2020, 70% | SSC, Lords Group of Schools, 2008–2018, 9.2 CGPA